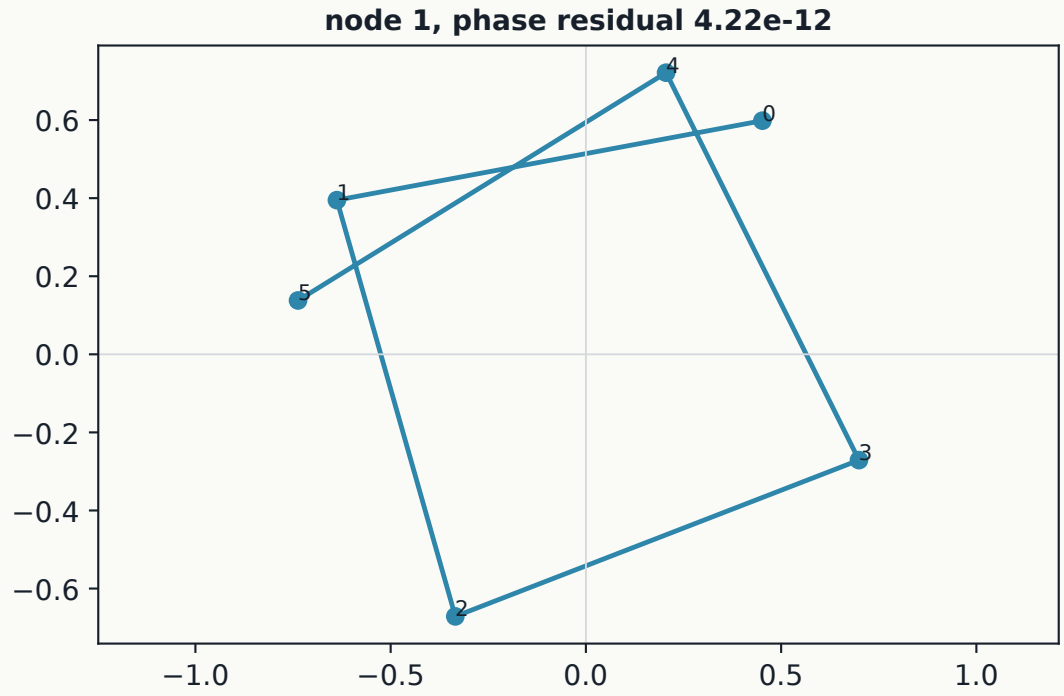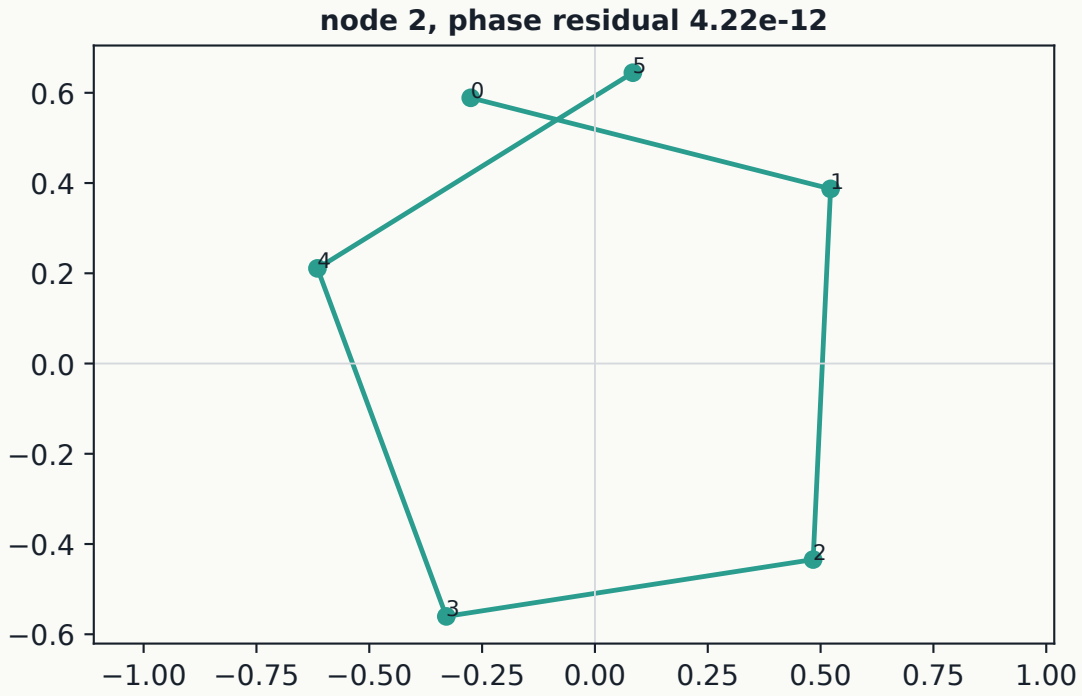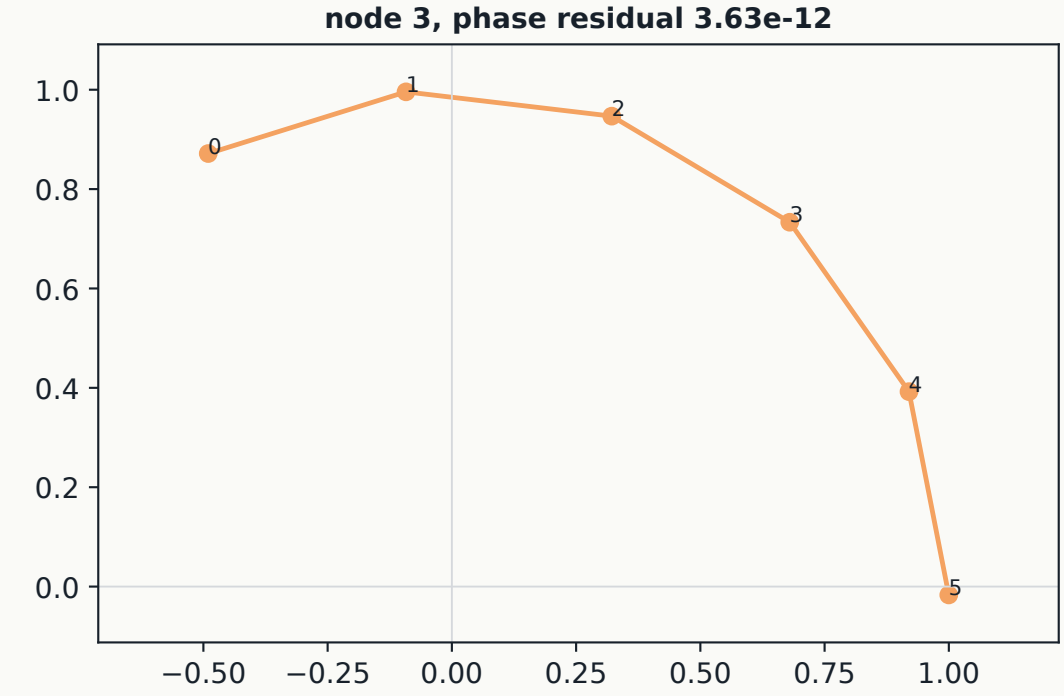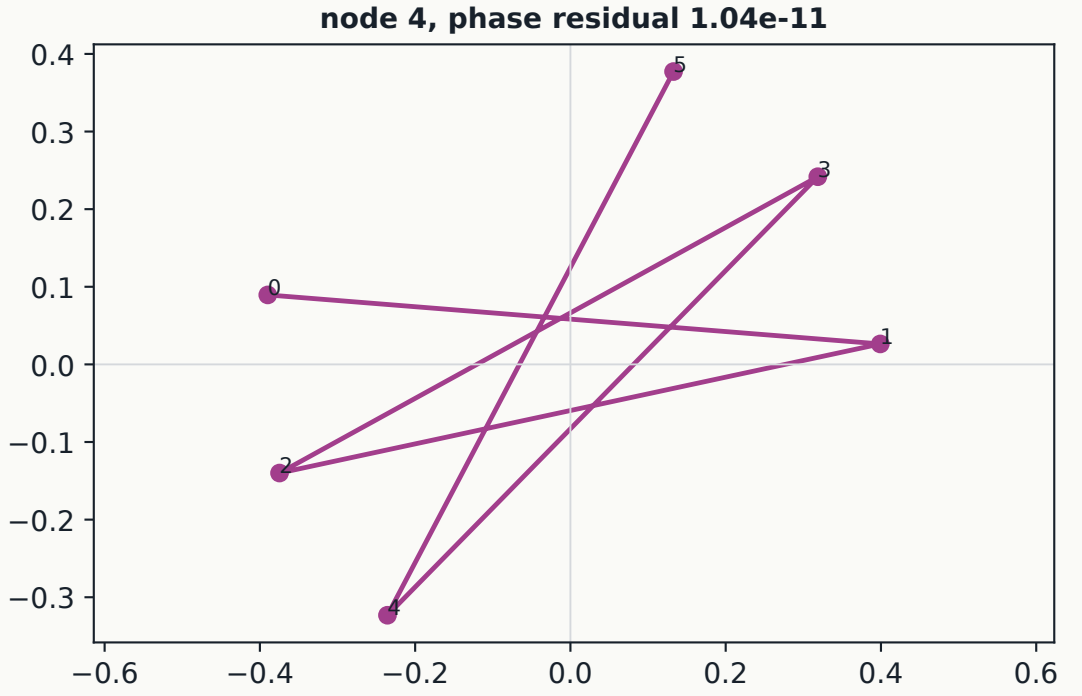Across offsets, each coefficient follows one geometric phase progression